The Limits of Educational Dynamics: Static vs. Dynamic Social Value Orientation in MARL

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Abstract

Standard reinforcement learning agents frequently struggle to achieve cooperation in complex environments characterized by competition and resource scarcity, commonly known as Sequential Social Dilemmas (7; 8). In such scenarios, the environmental payoff structure typically tempts agents towards selfish behavior, which ultimately degrades the collective welfare and leads to resource depletion. Human societies historically mitigate these dilemmas through the transmission of ideologies and values, rather than relying solely on pure self-interest. To model this socio-cultural phenomenon, we implement a dynamic education approach during training that shifts an agent's intrinsic goals over time using a Social Value Orientation (SVO) reward transformation. We trained Independent Proximal Policy Optimization (IPPO) agents within the DeepMind Melting Pot suite (8; 7) to evaluate if dynamic ideological curriculum could solve dilemmas that static reinforcement learning cannot. Our findings demonstrate that while a balanced cooperative ideology yields the most sustainable harvesting and highest collective reward, dynamically shifting this ideology during training introduces optimization instability that ultimately underperforms when compared to a static cooperative baseline. Videos of the simulations performed during our project can be found here: https://office365stanford-my.sharepoint.com/my?id=%2Fpersonal%2Flloydkam%5Fstanford%5Fedu%2FDocuments%2Fmeltingpot%5Fvids&viewid=59531657%2Db7a8%2D4ab9%2Dbb77%2Db5cadc28acae. Our GitHub repository can be found here: https://github.com/Lloydkamole/MARL_MeltingPot_cs234

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1. Introduction

When artificial agents interact in a shared, resource-constrained environment, they are frequently faced with situations where individual rationality leads to collective ruin. These environments are formalized as Sequential Social Dilemmas (SSDs) (7). Unlike simple matrix games such as the classic Prisoner's Dilemma, SSDs require agents to take a prolonged sequence of actions over time where the payoff structure heavily incentivizes a middle ground between selfish exploitation and cooperative restraint. If agents simply optimize for their raw, immediate environmental rewards, they almost inevitably fall into resources depletion. This results in suboptimal outcomes for the entire population, often culminating in the complete exhaustion of available resources. As artificial intelligence systems are increasingly deployed in real-world social environments—from autonomous driving networks to automated financial trading—their ability to successfully navigate coordination and complex resource negotiation becomes absolutely critical.

Human societies have developed sophisticated mechanisms to solve these exact types of resource dilemmas over thousands of years. This coordination is largely achieved through the dynamic transmission of cultural norms, structured education, and the adoption of shared ideologies (1; 6). In this project, we draw direct inspiration from these societal mechanisms in an attempt to improve multi-agent reinforcement learning (MARL). Instead of training agents on a static environmental reward signal, we model societal education by shifting an agent's intrinsic goals over time. We investigate if a simulated, dynamic education process can foster robust cooperative strategies that standard reinforcement learning algorithms fail to discover. This is accomplished by applying a Social Value Orientation (SVO) transformation to the rewards (3), which allows us to mathematically tune the balance between self-interest and the welfare of the broader group, gradually teaching the agents to value their peers.

2. Background and Related Work

The study of multi-agent reinforcement learning in sequential social dilemmas has expanded significantly as researchers seek to understand how cooperation emerges or collapses in decentralized systems. Previous literature highlights that standard algorithms like Q-learning or standard Policy Gradients often converge to aggressive, mutually destructive policies when deployed in environments with finite resources (7).

To rigorously evaluate our approach, we utilize the DeepMind Melting Pot simulator (2), specifically focusing on the Commons Harvest and Clean Up substrates (see Figure 1). Melting Pot was designed to provide a scalable and highly rigorous evaluation framework for MARL in complex, spatially extended environments. It requires agents to process raw pixel data and handle partial observability, closely mimicking the challenges of physical world deployment. Our core simulations consist of 1000 iteration steps using populations of 7 independent agents.

We mathematically model the underlying multi-agent problem as a Decentralized Partially Observable Markov Decision Process, or Dec-POMDP. This formal framework extends the traditional Markov Decision Process to accommodate multiple actors with limited information. It is defined by a specific tuple of elements. A population of N independent agents interacts within a true global state S. This global state encompasses the hidden grid of all positions, entities (like apples or waste), and environmental cooldowns. Because agents cannot see the entire world at once, an observation kernel O restricts the global state to local observations Z. For our trained agents, this observation manifests as an 88x88x3 RGB field of view, meaning they can only react to what is immediately in front of them.

At each timestep, the agents independently select actions from a discrete joint action space U. This space includes basic navigation (moving up, down, left, right), rotation, and specific interaction mechanics like zapping other agents or cleaning the environment. The simulator then updates according to transition dynamics P, which handle the physics of movements, collisions, and the respawning of resources. Following these environmental transitions, agents receive a team reward r that is discounted over time by a factor γ, forcing them to weigh immediate gratification against long-term survival.

3. Approach and Methodology

Our core methodology centers on intercepting the raw rewards provided by the Melting Pot environment and mathematically modifying them to reflect different social ideologies before they are passed to the learning algorithm.

3.1. Social Value Orientation Transformation

To model ideological beliefs accurately, we implemented a utility transformation wrapper based on the psychological concept of Social Value Orientation (3). This ensures that each agent's perceived reward is a calculated mixture of their own self-interest and the average reward of their surrounding peers. The modified utility $u_i(t)$ for agent i is calculated using a trigonometric formulation:

$$u_i(t) = \cos(\phi_i) \cdot r_i(t) + \sin(\phi_i) \cdot \overline{r}_{-i}(t) \tag{1}$$

In this equation, $r_i(t)$ represents the raw intrinsic reward given by the environment (e.g., eating an apple), and $\overline{r}_{-i}(t)$ is the mean reward achieved by all other agents in the simulation at that specific timestep. The parameter ϕ_i acts as the ideological dial for the agent. An angle of 0° represents pure selfishness, as the sine term cancels out the welfare of others entirely, and will serve as a baseline to benchmark our approach against. Conversely, an angle of $45°$ represents balanced cooperation, assigning equal weight to personal success and the success of the group. An angle of $90°$ simulates pure altruism, where the agent derives no pleasure from its own actions unless they benefit others. This custom wrapper successfully intercepts the environment's raw rewards and outputs the transformed values at every step of the simulation without breaking the underlying environment logic.

3.2. Curriculum Schedule Formulation

Motivated by coordination failures observed in early testing when agents were initialized with static altruistic ($\phi = 90°$) parameters, we implemented a dynamic curriculum state to explicitly model exogenous education. Our guiding hypothesis was that agents must first learn how to interact with the world and gather resources selfishly before they can effectively understand how to cooperate. To test this, we designed three distinct schedules for the ideological parameter ϕ, operating over the course of 100 training episodes.

The first approach is a **linear** curriculum, representing a steady and gradual integration of societal values. Over the training horizon, ϕ is increased at a constant rate from $\phi_{\min}$ to $\phi_{\max}$. For episode $k \in [0, N-1]$, the update rule is:

$$\phi_k = \phi_{\min} + (\phi_{\max} - \phi_{\min}) \cdot \frac{k}{N-1} \tag{2}$$

For the specific case used in your example ($N = 100$, $\phi_{\min} = 0°$, $\phi_{\max} = 90°$), this becomes:

$$\phi_k = 0° + (90° - 0°) \cdot \frac{k}{99} \tag{3}$$

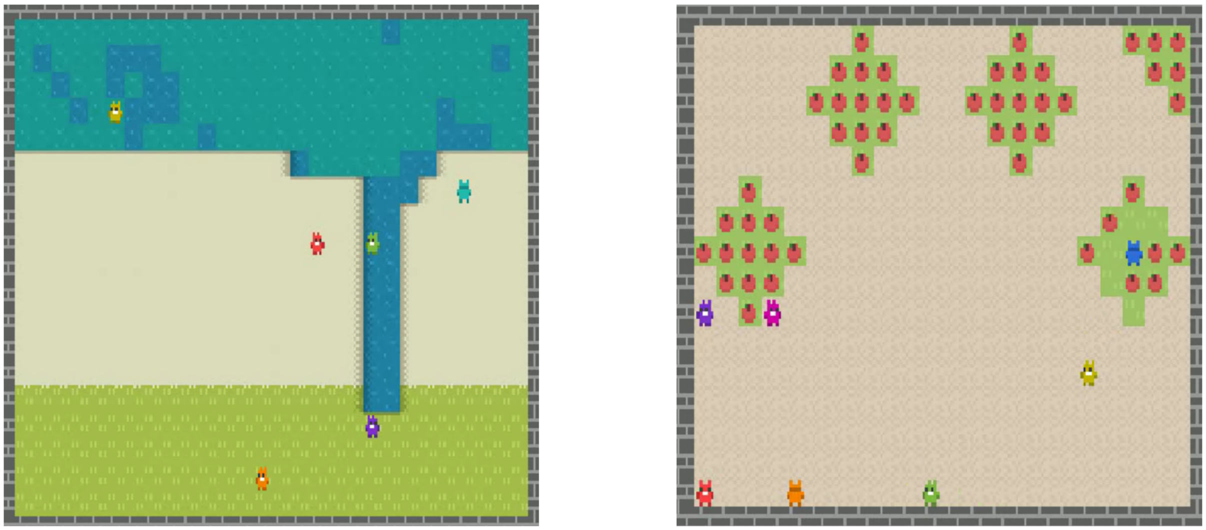

Figure 1. The two different substrates that were experimented during this project: Clean Up (right) and Common Harvest (left)

The second approach is a **logarithmic** curriculum. This models strong early social shaping that gradually tapers as agents mature. The normalized progress term is concave, causing faster growth early and slower growth later:

$$t_k = \frac{k}{N-1} \tag{4}$$

$$\phi_k = \phi_{\min} + (\phi_{\max} - \phi_{\min}) \ln(1 + (e-1)\, t_k) \tag{5}$$

The third approach is an **exponential** curriculum. This does the opposite: weak social pressure early, then progressively stronger pressure later. The normalized term is convex:

$$t_k = \frac{k}{N-1} \tag{6}$$

$$\phi_k = \phi_{\min} + (\phi_{\max} - \phi_{\min}) \cdot \frac{e^{t_k} - 1}{e - 1} \tag{7}$$

The fourth approach is a **sigmoid** curriculum, designed as an S-shaped transition: slow change at the beginning, rapid transition near the midpoint, and saturation near the end. Let

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \qquad m = \frac{N}{2} \tag{8}$$

Then define the normalized schedule term:

$$s_k = \frac{\sigma(k-m) - \sigma(-m)}{\sigma(N-1-m) - \sigma(-m)} \tag{9}$$

and the angle update:

$$\phi_k = \phi_{\min} + (\phi_{\max} - \phi_{\min})\, s_k \tag{10}$$

In our current setup, all agents share the same curriculum value each episode, so for agent i:

$$\phi_k^{(i)} = \phi_k \tag{11}$$

For all schedules, each episode initializes a fresh environment wrapped with our transformation, ensuring all agents undergo identical, synchronized ideological shifts. This setup allows us to rigorously test whether gradual, rapid, or punctuated educational environments lead to more robust cooperative policies.

3.3. Agent Architecture and Policy Optimization

Replacing basic random action policies, we integrated trained Independent Proximal Policy Optimization (IPPO) agents. Using IPPO allows each agent to treat the others as part of their own non-stationary environment. This decentralized approach is computationally heavier but much more accurately simulates human interaction, where individuals do not share a hive-mind policy network.

Because the Melting Pot environments feature strict partial observability, agents without memory cannot function properly. They must remember past interactions to build trust or hold grudges. We resolve this by implementing a neural network architecture that begins with a Convolutional Neural Network (CNN) to process the local RGB pixel observations. The flattened output is then passed through fully connected layers and fed into a Long Short-Term Memory (LSTM) network. This recurrent memory helps agents understand the long-term consequences of their actions and track the behavior of their peers even when those peers leave their immediate field of view. Policy updates are stabilized using PPO-Clip, ensuring that the agents do not completely forget their previous behaviors when exposed to new situations, and action valuation is balanced using Generalized Advantage Estimation.

4. Experimental Results

To quantitatively evaluate our hypotheses without interrupting the environment execution, we built a robust, real-time metrics pipeline. Throughout our experiments, we focused our analysis on two primary metrics. First, Mean Total Reward tracks the cumulative environmental reward harvested by the population, representing utilitarian welfare. Second, the Gini Coefficient measures the inequality of those returns across the agent population, tracking how fairly the wealth is distributed among individuals.

4.1. Training Dynamics

During the training phase, we monitored how different static SVO parameters influenced the learning curves. Figure 2 illustrates the progression of the reward mean, the Gini coefficient, and episode lengths over millions of environment steps.

The training data (Figure 2) clearly shows that the $45°$ cooperative agents (orange line) learn the most effective policies, achieving a rapidly climbing reward mean that surpasses all other configurations. In contrast, the purely selfish $0°$ agents (blue line) learn much slower and converge at a significantly lower mean reward. The subfigure in the middle of Figure 2 is particularly telling: selfish agents experience a massive spike in the Gini coefficient during early training, which remains elevated, indicating that only a few aggressive agents are successfully learning to hoard resources. The purely altruistic $90°$ agents (red line) completely fail to learn, as their reward mean flatlines almost immediately due to extreme passivity.

To better visualize the asymptotic performance of these policies, we applied a rolling mean (window size of 5) to the reward signals, filtering out the high step-to-step variance inherent in multi-agent environments.

As shown in Figure 3, this smoothed convergence trajectory highlights the performance ceilings of each ideology. The $45°$ balanced cooperative policy demonstrates continuous, stable optimization, ultimately converging near a peak reward of 500. The $70°$ policy, while cooperative, plateaus noticeably lower around 450. Conversely, the purely selfish $0°$ policy converges prematurely around a ceiling of 360, unable to unlock the higher collective rewards that require coordination and resource preservation. Finally, the $90°$ altruistic agents show almost no learning curve, flatlining near 230, demonstrating that lacking self-interest entirely prevents the acquisition of basic harvesting competencies.

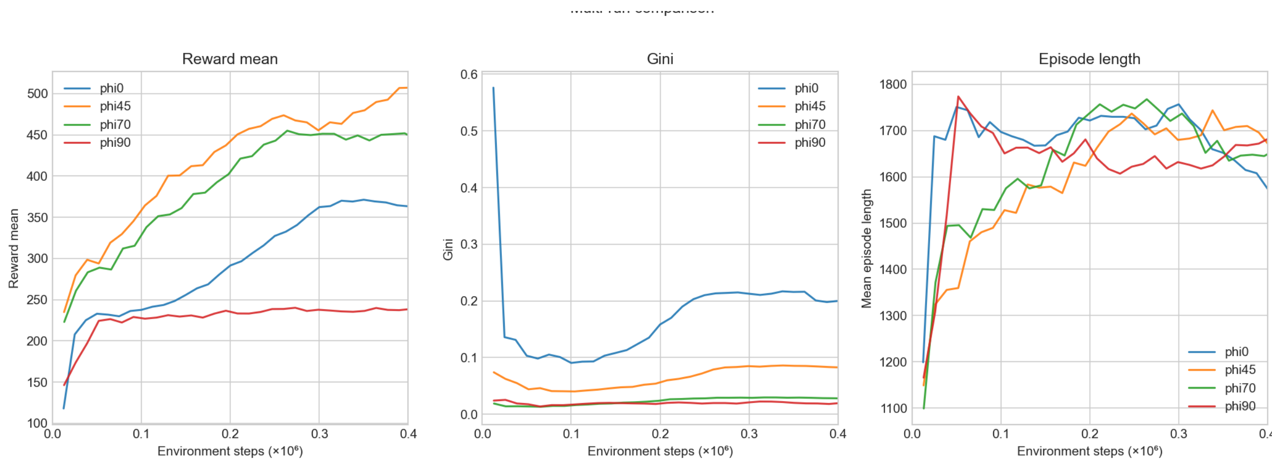

Figure 2. Training metrics across environment steps for different static ϕ values. The 45° population quickly dominates in mean reward while maintaining low inequality.

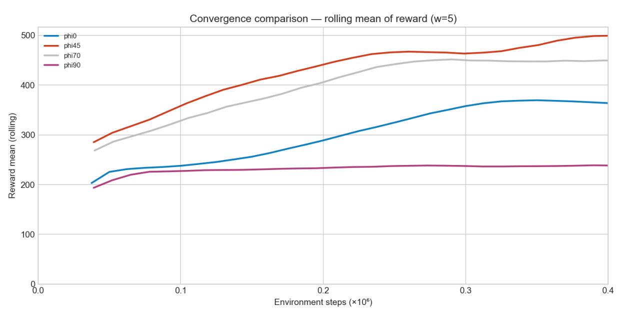

Figure 3. Convergence comparison showing the rolling mean of the reward (window=5) across environment steps.

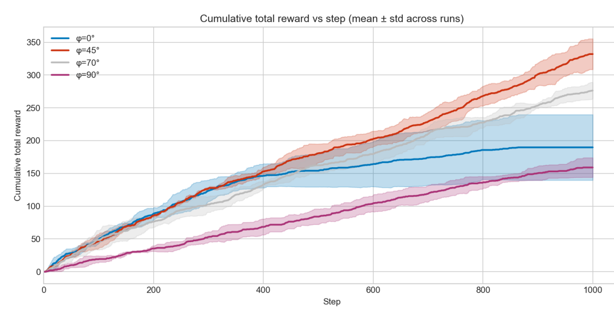

Figure 4. Cumulative total reward vs step (mean $\pm$ std across runs) in the Commons Harvest simulation. The selfish 0° policy plateaus as resources are exhausted.

4.2. Evaluation of Static Ideologies in Simulation

Following training, we evaluated the converged policies in 1000-step simulations (average simulation duration of a substrate according to the creators of MeltingPot). The variations in ϕ produced drastically different, and sometimes fatal, societal outcomes across the simulated populations.

As depicted in Figure 4, the simulated 0° population suffers from non sustainable behavior induced by the selfish reward function during the training. The cumulative reward for these selfish agents completely plateaus around step 400. They aggressively rush the resource spawns, exhausting the apples completely. Because no apples are left to trigger the regrowth mechanic, the environment becomes barren, and the agents effectively starve for the remaining 600 steps.

Conversely, the 45° agents demonstrate sustainable harvesting. The cumulative reward curve climbs steadily for the entire 1000-step duration. These cooperative agents learned to leave one or two apples remaining in a cluster to allow the crops to replenish themselves, securing a long-term food supply.

To summarize the trade-offs between overall welfare and societal fairness, we analyzed the normalized multi-metric scores shown in Figure 5. The 45° ideology perfectly maximizes both the total normalized reward and equality (where high equality represents a low Gini coefficient). While a 70° ideology maintains a decent total reward, it sacrifices equality. The 0° ideology fails dramatically on both fronts, proving that pure self-interest in an SSD degrades both individual and collective success.

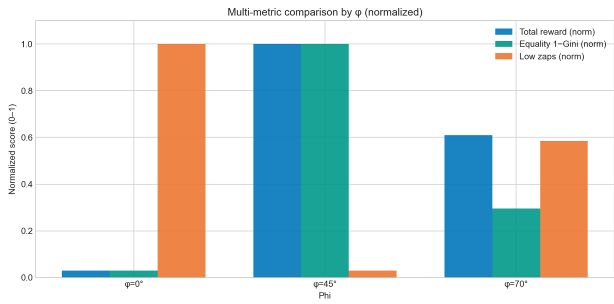

Figure 5. Normalized multi-metric comparison showcasing the balance of Total Reward and Equality (calculated as 1 - Gini) across ϕ configurations.

4.3. Curriculum Schedule Performance

After establishing the clear superiority of the $45°$ static baseline, we tested our dynamic curriculum schedules. We strongly hypothesized that allowing agents to first learn basic task competence while selfish, and then gradually educating them towards cooperation via linear, logarithmic, or sigmoid schedules, would yield superior and more robust policies.

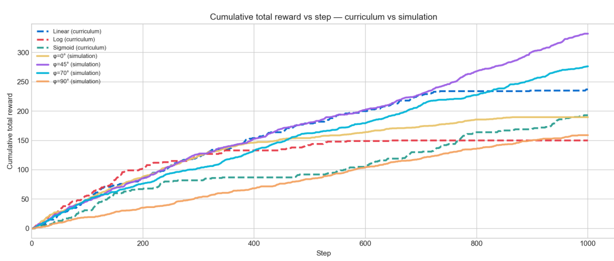

Figure 6. Cumulative total reward vs step comparing the linear, log, and sigmoid curriculum schedules against the static ϕ baselines.

However, the experimental data thoroughly contradicted our hypothesis. As shown in the temporal tracking of Figure 6, the linear, logarithmic, and sigmoid curricula start off pacing the selfish agents but fail to ever catch up to the trajectory of the static $45°$ baseline.

Particularly notable is the failure of the **Logarithmic Curriculum**. Because the log function imposes a rapid early shift away from selfishness, it performed the worst among the dynamic schedules, severely plateauing around a cumulative reward of 150. This demonstrates that forcing agents out of the selfish exploration phase too quickly stunts their ability to learn fundamental environmental competencies, trapping them in passive inefficiency before they even figure out how to harvest.

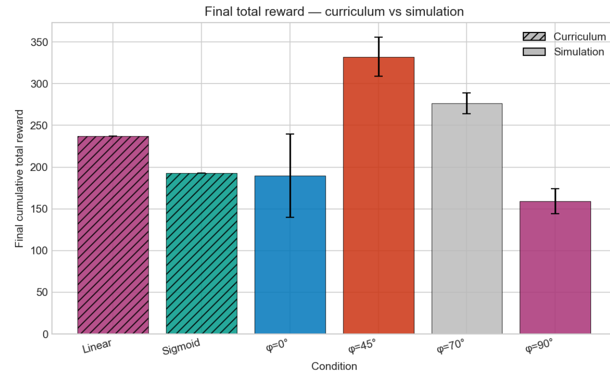

Figure 7. Final cumulative total reward across all experimental conditions, complete with error bars.

Figure 7 provides the definitive summary of these outcomes. The dynamic curriculum schedules consistently failed to outperform the static baselines. The curriculum schedules yielded final total rewards that fell well short of the impressive heights achieved by the static $45°$ baseline.

Our analysis indicates that dynamically shifting the SVO ideology mid-episode introduces severe optimization instability. Because the intrinsic reward function effectively acts as a moving target, the IPPO agents' advantage estimations become deeply skewed, and the network struggles to converge on a stable policy. What was considered a highly rewarding action at episode 10 is suddenly punished or ignored at episode 50. Ultimately, the agents achieve the highest utilitarian welfare when they are initialized with a fixed, optimal cooperative ideology from step zero, allowing the neural network to map state-action pairs to a mathematically consistent objective.

5. Conclusion and Future Work

This research successfully implemented a Social Value Orientation reward wrapper to mathematically model ideological constraints in complex sequential social dilemmas. Our empirical evaluations within the Melting Pot suite revealed that an ideology tuned to $45°$ yields the most optimal agent policy. Balanced cooperation returns the highest amount of cumulative reward and successfully enforces sustainable harvesting behaviors that prevent the resource depletion, all while maintaining a remarkably low Gini coefficient. However, we discovered that dynamic curriculum schedules introduce too much variance into the learning process. The shifting reward horizons prevented the agents from learning as effectively as those anchored to a static optimal ideology.

Moving forward, several promising avenues remain for extending this work. First, we plan to investigate the impact of alternative wrappers on agent policies, specifically targeting the Clean Up substrate (see Figure 1). Clean Up is a game where agents must choose between taking apples now for quick reward, or cleaning the river so apples keep growing later, with reward averaging around closer to 0. If nobody cleans, the environment gets dirty and apple growth drops for everyone. So it tests whether agents can cooperate for long-term benefit. Preliminary results using our SVO wrapper for training have shown that our training methodology isn't yet adequate to reach optimal agent policies on Clean Up. The agents don't seem to understand that they need to clean the river to be able to start collecting apples. That is why, for tasks that require dedicated labor rather than merely resource restraint (e.g., Clean Up vs Common Harvest), we propose developing a secondary wrapper based on the Fehr-Schmidt Inequity Aversion model (4). This approach could penalize deviations from equality via envy (α) and guilt (β) parameters, computing per-agent utility as:

$$u_i = r_i - \frac{\alpha_i}{N-1} \sum_{j \neq i} \max(r_j - r_i, 0) - \frac{\beta_i}{N-1} \sum_{j \neq i} \max(r_i - r_j, 0) \tag{12}$$

Furthermore, we aim to explore the concept of Ideological Contagion through decentralized viral education (5). Instead of forcing a globally synchronized schedule upon the agents, the agents could dynamically update their internal ϕ based on perceived success within their local observation kernel. If an agent observes a neighbor performing exceptionally well, they would shift their own goals to match that neighbor's ideology. This would create a much more organic, decentralized social influence model, potentially allowing cooperative behavior to spread naturally through a population without destabilizing the core optimization process. The videos of the simulations performed during our project can be found here: https://office365stanford-my.sharepoint.com/my?id=%2Fpersonal%2Flloydkam%5Fstanford%5Fedu%2FDocuments%2Fmeltingpot%5Fvids&viewid=59531657%2Db7a8%2D4ab9%2Dbb77%2Db5cadc28acae. The link to ur GitHub repository can be found here: https://github.com/Lloydkamole/MARL_MeltingPot_cs234

Statement of Contributions

Both authors contributed equally to the conceptualization, implementation, and evaluation of this research. Nael Ghoundale focused extensively on the environment integration with the Melting Pot suite and architected the real-time metrics pipeline. Lloyd Kamole Ghomsi managed the formulation of the IPPO agent architecture, the LSTM memory integration, and the mathematical logic for the curriculum schedules. Both members collaborated closely on data analysis and the drafting of the final manuscript.

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